

JORDAN NEWMAN

Systems Software Engineer | Kernel, Networking, Embedded Firmware

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SUMMARY

Systems software engineer with deep C, FreeBSD, Linux, networking, security, kernel, and embedded firmware experience. Built production software from kernel modules and packet analyzers to high-performance web servers and embedded AVR firmware. Operated infrastructure at ISP/CDN scale across 3,000-4,000 servers, 10 CDN locations, and 65+ Gbps peak traffic.

TECHNICAL SKILLS

Languages: C, C++, Go, Python, C#, Objective-C, PHP, Perl, JavaScript, Node.js, Shell, Visual Basic

Systems: FreeBSD, Linux, kernel modules, system calls, bootloaders, operating systems, kqueue, GDB, TCP/IP

Networking/Security: DDoS detection/mitigation, packet capture, DNS analysis, libpcap, IPFW, authentication, telemetry

Embedded: Embedded C, Atmel AVR, SPI, UART, I2C, Arduino, circuit design, soldering, hardware troubleshooting

Cloud/DevOps: Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, security scanning, deployment automation

Databases/Web: MySQL, MultiValue/Universe, OAuth/OAuth2, SMTP OAuth, React, jQuery, PHP, web services

SELECTED TECHNICAL HIGHLIGHTS

- Built and operated infrastructure spanning 3,000-4,000 servers, 10 CDN locations, and 65+ Gbps peak traffic.
- Developed PrimeHTTPD in C around FreeBSD kqueue; supported 150,000+ concurrent connections across ~200 production servers.
- Created FreeBSD kernel modules for security, filesystem virtualization, and filesystem observability in customer-facing production systems.
- Developed libpcap-based network and DNS analyzers for real-time traffic visibility, DDoS analysis, and DNS recursion attack detection.
- Developed embedded C firmware for battery-powered wireless Atmel AVR smart smoke/CO2 detector prototypes using SPI, UART, and I2C.
- Built custom x86 OS kernel and bootloader with protected mode, IDT, heap allocation, port I/O, VGA output, and GDB debugging.

SELECTED SYSTEMS PROJECTS

TAFOS Operating System: Developed educational x86 OS kernel in C and assembly with custom MBR bootloader, protected mode, IDT, heap allocator, port I/O, VGA output, and GDB support.

KeepClean: Developed FreeBSD kernel module hooking syscalls to monitor execution activity, protect system assets, and report telemetry to userspace services.

VirtualDir: Developed FreeBSD kernel module remapping filesystem paths through syscall interception; deployed customer-facing across approximately 30 production servers over 5-8 years.

StatCache: Developed FreeBSD kernel module instrumenting stat() and lstat() activity for filesystem visibility and performance analysis.

PrimeDump: Developed libpcap-based traffic analyzer decoding Ethernet, IP, TCP, UDP, and ICMPv6 headers with ncurses real-time views and IPFW integration.

PrimeDNSTop: Developed DNS traffic monitor parsing UDP/53 and RFC 1035 queries to detect recursion attacks and rank source/domain activity.

AVR Smart Smoke/CO2 Detector: Developed contract firmware in embedded C for battery-powered wireless Atmel AVR prototypes using SPI, UART, and I2C interfaces.

PROFESSIONAL EXPERIENCE

Advantive - Senior Development Specialist | Remote / Tampa Bay, FL | July 2022 - Present

- Implemented hands-on Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, and security-scanning workflows across engineering teams.
- Partnered with architecture teams to improve deployment reliability, DevOps policies, and production software delivery practices.

DDI Systems - Senior Development Specialist | Manalapan, NJ | November 2021 - July 2022

- Modernized legacy VB.NET software by porting components to C#, replacing deprecated and insecure implementations.

ISPrime LLC - CEO | Weehawken, NJ | April 2018 - January 2020

- Led datacenter modernization, operational cleanup, and software platform improvements for production hosting infrastructure.

DDI Systems - Computer Programmer | Manalapan, NJ | April 2017 - April 2018

- Integrated ERP software with SellerCloud and Essendant systems, automating order, catalog, image, and SDS data workflows.

MFCXY, Inc. - Computer Programmer | Chicago, IL | December 2014 - September 2015

- Developed backend server features and Windows client integrations while identifying performance and security issues.

ISPrime Inc. - CIO / Partner | Weehawken, NJ | March 2001 - December 2014

- Built and operated infrastructure spanning 3,000-4,000 servers across 10 CDN locations and multiple datacenters.

- Sustained peak traffic exceeding 65 Gbps while supporting mission-critical customer workloads.
- Developed PrimeHTTPD, a high-performance C web server optimized around FreeBSD kqueue that supported 150,000+ concurrent connections and demonstrated performance competitive with, and in some workloads exceeding, nginx.
- Deployed PrimeHTTPD across approximately 200 production servers supporting high-traffic customer environments.
- Developed DDoS detection and mitigation systems protecting infrastructure during attacks involving up to 750,000 compromised hosts.
- Created PrimeDump and PrimeDNSTop, libpcap-based network analysis tools for traffic anomalies and DNS attacks.
- Developed production FreeBSD kernel modules including KeepClean, VirtualDir, and StatCache for security and filesystem observability.
- Built monitoring, telemetry, authentication, and security systems used across thousands of production servers.

FatWallet.com - Systems Administrator | *Wisconsin / Remote* | *December 2000 - June 2001*

- Rebuilt production infrastructure to resolve performance and stability issues without customer-facing downtime.

EDUCATION

Brookdale Community College - Computer Science coursework; left during first semester to pursue ISPrime full-time during rapid company growth.